



THERMAL + VISUAL PERIMETER SECURITY / DAHEJ LNG TERMINAL

Seeing Beyond the Visible

Bispectral surveillance engineered to detect, verify, track and support response across a critical energy environment.



DOCUMENTED SCOPE

2 thermal PTZ units

AXIS PLATFORM

Q8742-LE

PROJECT LOCATION

Dahej, Gujarat

Detect early. Verify visually. Respond with confidence.

Thermal contrast for detection, plus HDTV visual imaging for verification. Petronet LNG Purchase Order 4500012204, dated 23 February 2021.

PROJECT BRIEF

Security mandate at a glance.

Petronet LNG placed an order for two AXIS Q8742-LE bispectral thermal PTZ network cameras for critical outdoor zones at its Dahej LNG Terminal.



THE REQUIREMENT

Thermal CCTV coverage for critical outdoor zones.

A coordinated thermal and visual view helps operators detect activity through heat contrast and assess it with a visible image and PTZ positioning.

ORDER REFERENCE

4500012204

ORDER DATE

23 Feb 2021

DELIVERY TARGET

31 May 2021

SECURITY OBJECTIVE

Strengthen long-range situational awareness with a dual-sensor system that can first detect activity through thermal contrast, then use the visual channel and PTZ control for faster assessment.

CLIENT-SAFE PROJECT FACTS

Scope: Procurement of thermal CCTV cameras | Quantity: 2 units
Commercial values, tax data, personal contacts and signatures are excluded.

RISK CONTEXT

Why LNG security needs more than normal CCTV.

Large energy terminals combine extended perimeters, open process areas, marine exposure and round-the-clock operations. Thermal imaging adds a detection layer based on heat contrast.

01

Perimeter intrusion

Detect people or vehicles where distance, darkness or background camouflage reduce visual confidence.

02

Zero and low light

Maintain detection capability when visible illumination is limited or absent.

03

Challenging atmosphere

Improve awareness where haze, glare, shadows or mixed lighting complicate visual-only monitoring.

04

Wide outdoor zones

Use endless pan and presets to inspect multiple areas from a controlled position.

05

Unattended movement

Use analytics, scheduled events and guard tours to support proactive monitoring.

06

Abnormal heat events

Use temperature-related event triggers as an operational alert input, subject to configuration and validation.

IMPORTANT SCOPE NOTE

This case study does not claim explosion-proof certification or replacement of an automatic fire-alarm system. Those applications require separate engineering validation.

AXIS TECHNOLOGY

AXIS Q8742-LE bispectral thermal PTZ.

Two sensors provide a coordinated view: thermal detection with visible video for assessment and verification.

BISPECTRAL THERMAL PTZ



MODEL Q8742-LE

THERMAL SENSOR

640 × 480 uncooled microbolometer

VISIBLE VIDEO

Up to HDTV 1080p

VISUAL ZOOM

30× optical + 12× digital

PTZ MOVEMENT

360° endless pan; −90° to +45° tilt

THERMAL OPTICS

35 mm lens; 17° horizontal field

SENSITIVITY

NETD below 50 mK

ENVIRONMENTAL RATING

IP66 + NEMA 4X

OPERATING RANGE

−50°C to +55°C

ENVIRONMENT

Marine suitable

INTEGRATED INTELLIGENCE AND EVENT HANDLING

Analytics

Video Motion Detection, Perimeter Defender and Axis camera applications.

Event actions

Pre/post-alarm buffering, notifications, image/video upload, guard tours and outputs.

Event triggers

Analytics, temperature, inputs, schedules, edge-storage events and PTZ presets.

Recording options

SD-card edge recording and network-share recording.

OPERATIONAL VALUE

From movement to an actionable security event.

Thermal and visible channels work as complementary layers, helping operators move from detection to assessment and a documented response.

01

DETECT

Thermal contrast reveals activity independent of visible colour or normal lighting.

02

VERIFY

The visible HDTV channel and IR illumination support rapid scene assessment.

03

TRACK

Responsive PTZ, presets and guard tours help operators maintain situational focus.

04

ALERT

Analytics and configurable events can create notifications or initiate recording actions.

05

RECORD

Edge and network recording options preserve pre-alarm, event and post-alarm evidence.

06

RESPOND

Operators gain earlier context for escalation, patrol direction and incident handling.

SECURITY PRINCIPLE

Thermal finds the anomaly. Visual video provides the context. The operator makes the decision.

THERMAL IMAGING

Seeing heat contrast across the scene.

The comparison uses the supplied LNG terminal image to explain the operating concept. The false-colour panel is illustrative and is not a camera capture or temperature measurement.



VISIBLE-LIGHT REFERENCE



ILLUSTRATIVE THERMAL VIEW

WHAT THERMAL ADDS TO THE OPERATOR'S VIEW

Contrast over colour

Thermal sensing responds to temperature differences rather than surface colour.

Faster anomaly search

Heat contrast can help locate activity in complex outdoor scenes.

Day/night continuity

Detection is less dependent on visible illumination and shadows.

Paired verification

The visual sensor supports identification and contextual confirmation.

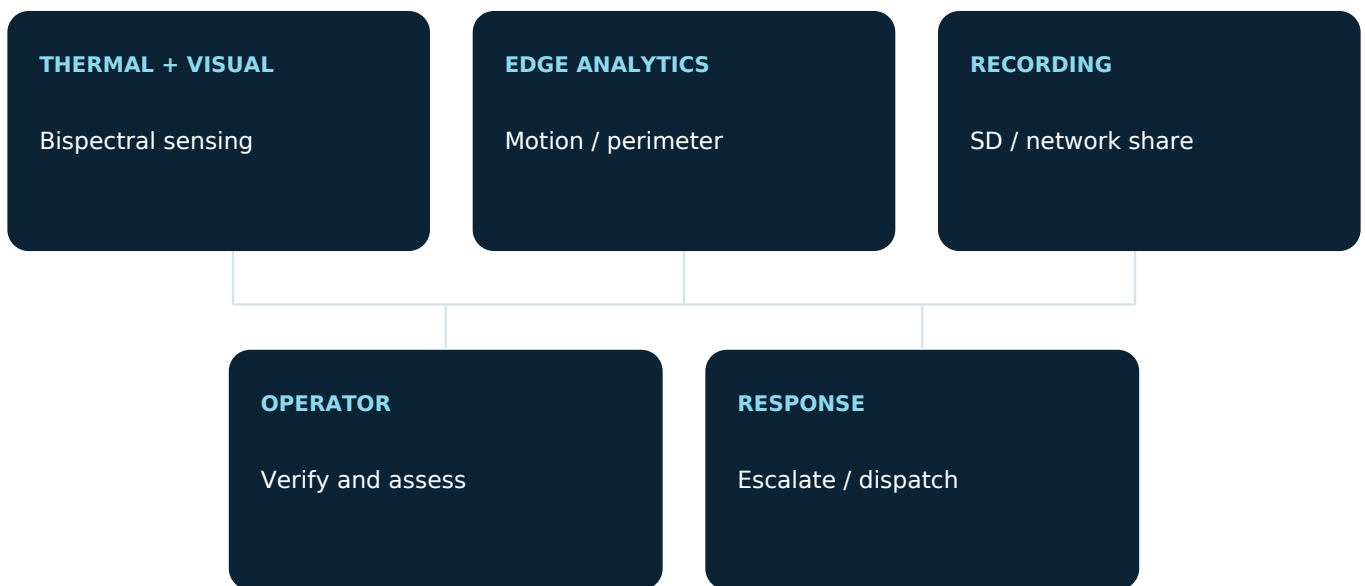
ILLUSTRATION DISCLAIMER

Applied colours do not represent calibrated temperatures or recorded alarm evidence.

SECURITY ARCHITECTURE

Camera-to-response workflow.

The camera supports network streaming, edge storage, analytics and event actions. Its value grows when events are linked to recording, operator verification and response procedures.



OPERATIONAL CONTROLS

Guard tours and presets

Create repeatable observation patterns for defined zones.

Pre/post-alarm buffering

Retain context before and after a trigger, subject to configuration.

Secure network transport

The specification includes HTTPS, SFTP and network-share event actions.

Environmental resilience

Outdoor construction, wide temperature range and marine suitability support terminal conditions.

CYBERSECURITY AND LIFECYCLE NOTE

The model reflects the 2021 project scope. Review firmware, credentials, network segmentation and lifecycle status periodically against current Axis guidance and the customer's security policy.

PROJECT PROOF

Critical infrastructure security outcome.

A documented two-camera deployment at Petronet LNG, Dahej, brings thermal detection and visible verification into one outdoor security workflow.



PETRONET LNG / DAHEJ LNG TERMINAL

A security layer built to detect what ordinary CCTV may miss.

Two documented AXIS bispectral PTZ cameras combine thermal detection, visible verification, responsive positioning, analytics and rugged outdoor operation.

DOCUMENTED QUANTITY

2 units

SECURITY METHOD

Detect + verify

ENVIRONMENT

Outdoor / marine

WHY THIS CASE STUDY MATTERS TO A NEW CLIENT

Verified enterprise deployment

The formal purchase order supports the project reference.

Purpose-fit camera selection

Bispectral PTZ was selected for long-range, all-light situational awareness.

Operational integration

Analytics, recording and event actions support a response workflow.

Industrial environment readiness

Specified casing and operating limits address demanding outdoor conditions.

EVIDENCE AND SOURCES

Petronet LNG PO 4500012204 and Annexure-B; AXIS Q8742-LE product datasheet and support documentation.